

Lesson 2 Data Visualization Principles

Libraries

```
library(tidyverse)
```

Data import

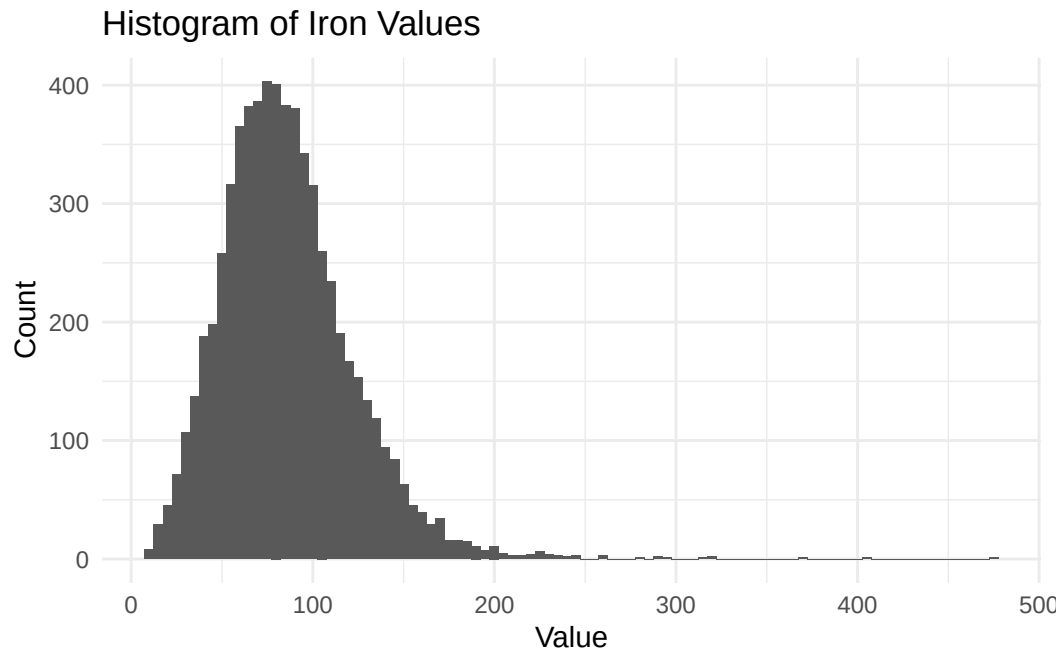
```
chem <- read_csv("data/chem_data.csv")
chem_wide <- chem |>
  pivot_wider(names_from = test, values_from = value)

iron <- chem |>
  filter(test == "iron")

exp <- read_csv("data/lab_exp.csv")
exp_long <- exp |> pivot_longer(personnel:total,
                               names_to = "type",
                               values_to = "cost")
```

Histogram

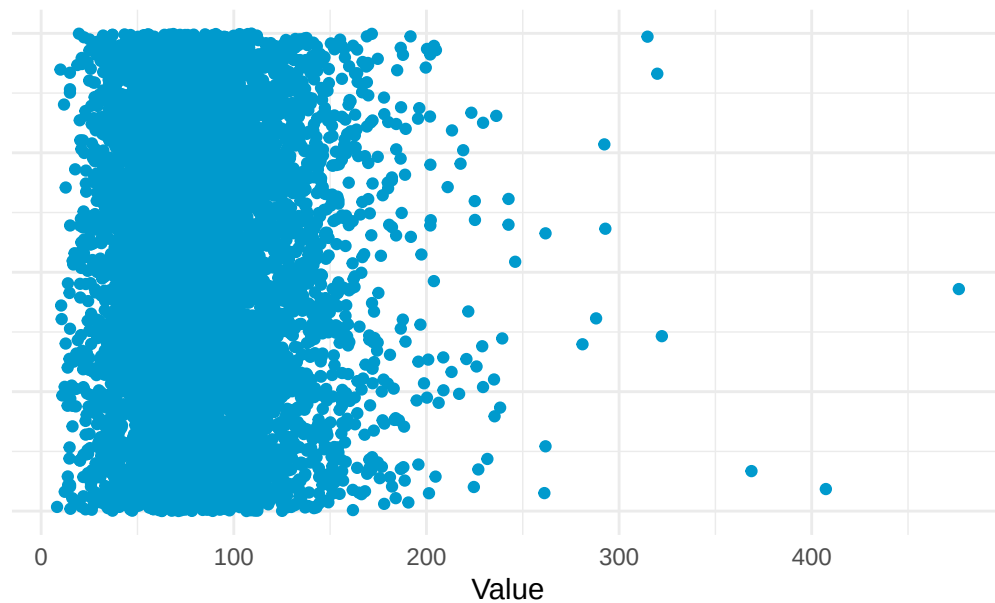
```
ggplot(data = iron, mapping = aes(x = value)) +
  geom_histogram(binwidth = 5) +
  theme_minimal() +
  labs(x = "Value", y = "Count", title = "Histogram of Iron Values")
```



Strip plot

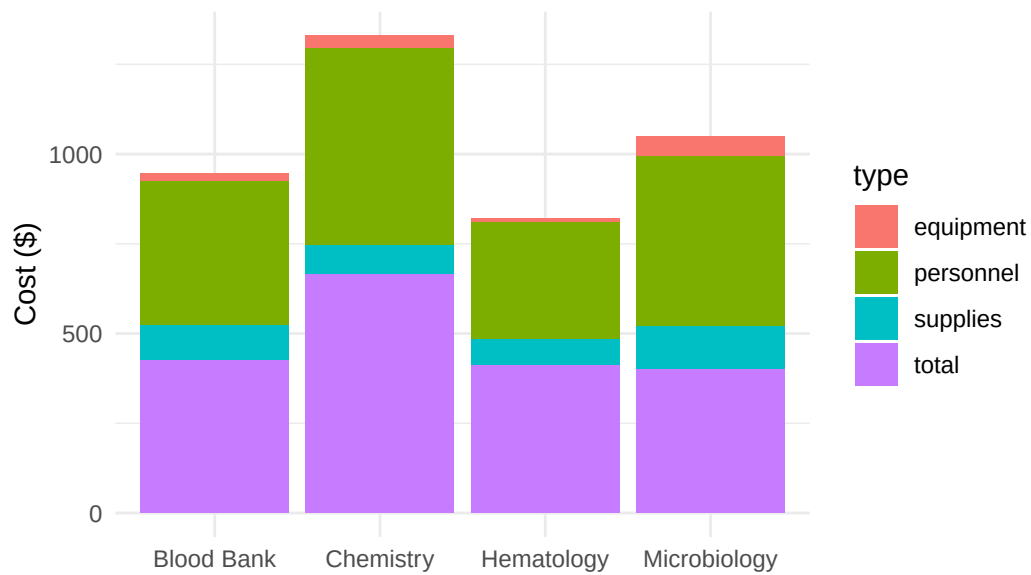
```
ggplot(data = iron, mapping = aes(x = value, y = 1)) +  
  geom_jitter(color = "deepskyblue3") +  
  theme_minimal() +  
  theme(axis.text.y = element_blank()) +  
  labs(x = "Value", y = "", title = "Strip Plot of Iron Values")
```

Strip Plot of Iron Values

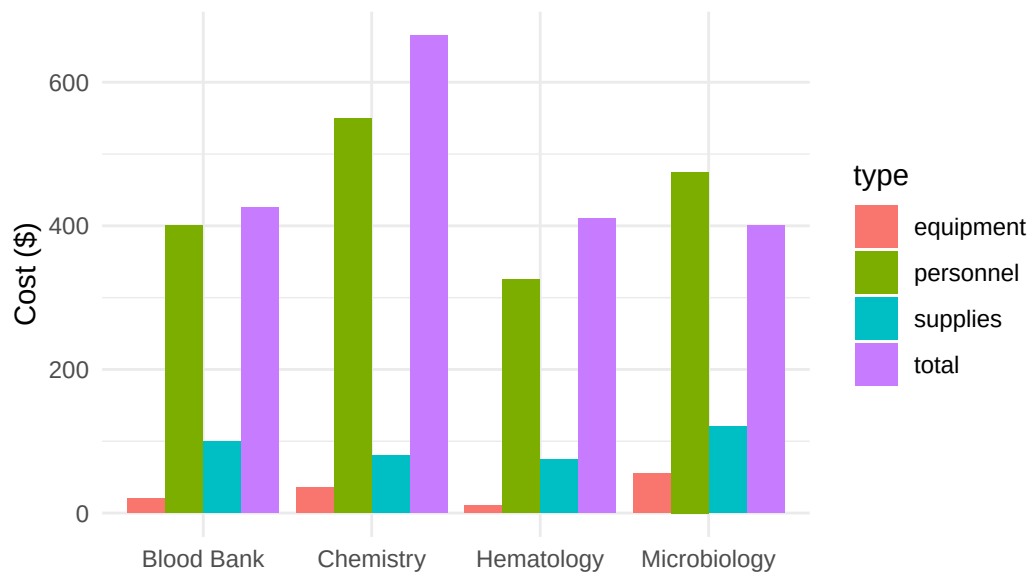


Bar plots

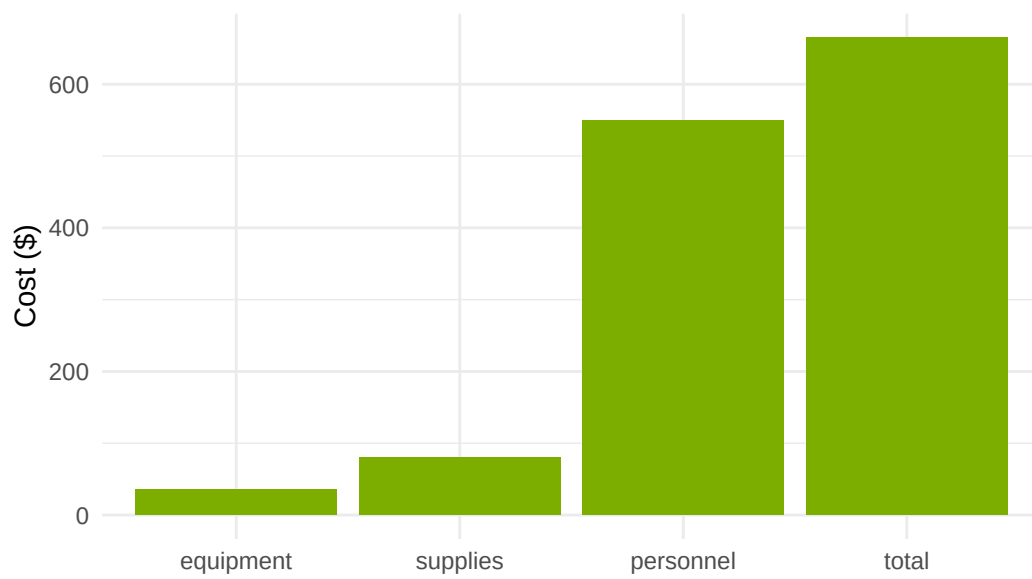
```
ggplot(data = filter(exp_long,  
  department %in% c("Chemistry", "Blood Bank",  
    "Microbiology", "Hematology"))) +  
  geom_bar(mapping = aes(x = department,  
    y = cost,  
    fill = type),  
    stat = "identity") +  
  theme_minimal() +  
  labs(x = "", y = "Cost ($)",  
    title = "")
```



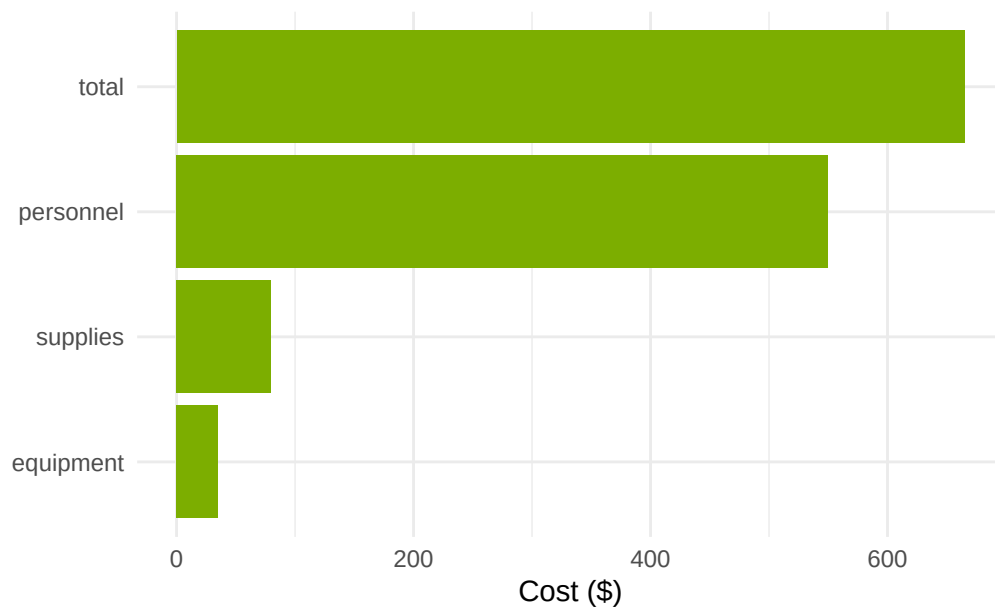
```
ggplot(data = filter(exp_long,
  department %in% c("Chemistry", "Blood Bank",
    "Microbiology", "Hematology"))) +
  geom_bar(mapping = aes(x = department,
    y = cost,
    fill = type),
    stat = "identity",
    position = "dodge") +
  theme_minimal() +
  labs(x = "", y = "Cost ($)",
    title = "")
```



```
ggplot(data = filter(exp_long, department == "Chemistry")) +  
  geom_bar(mapping = aes(x = fct_reorder(type, cost),  
                           y = cost),  
           stat = "identity",  
           fill = "#7CAE00") +  
  theme_minimal() +  
  labs(x = "", y = "Cost ($)",  
       title = "")
```

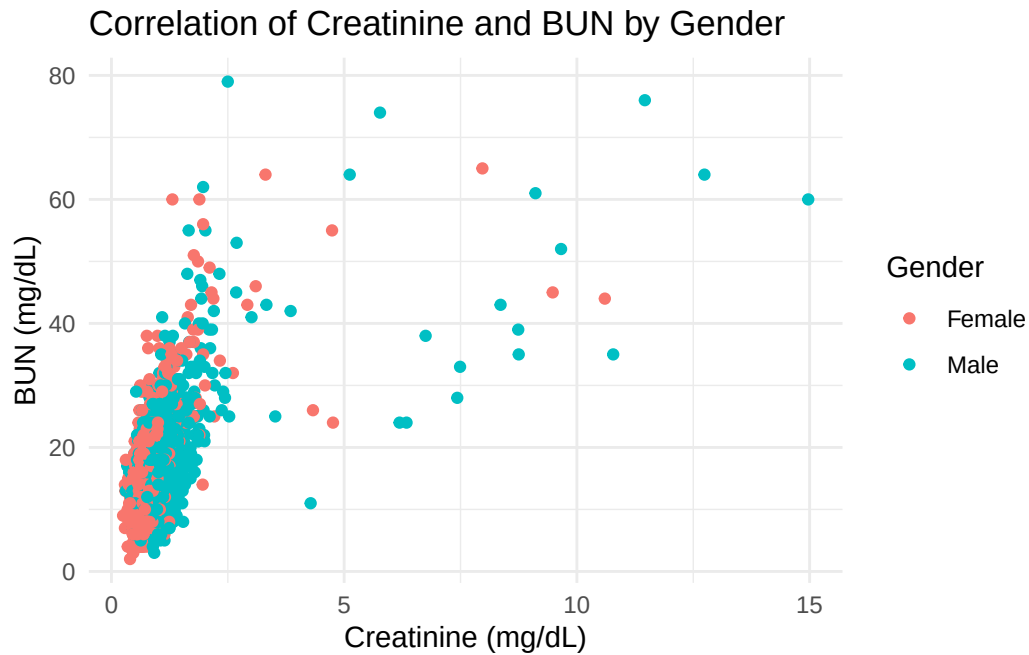


```
ggplot(data = filter(exp_long, department == "Chemistry")) +  
  geom_bar(mapping = aes(x = fct_reorder(type, cost),  
                             y = cost),  
           stat = "identity",  
           fill = "#7CAE00") +  
  coord_flip() +  
  theme_minimal() +  
  labs(x = "", y = "Cost ($)",  
       title = "")
```



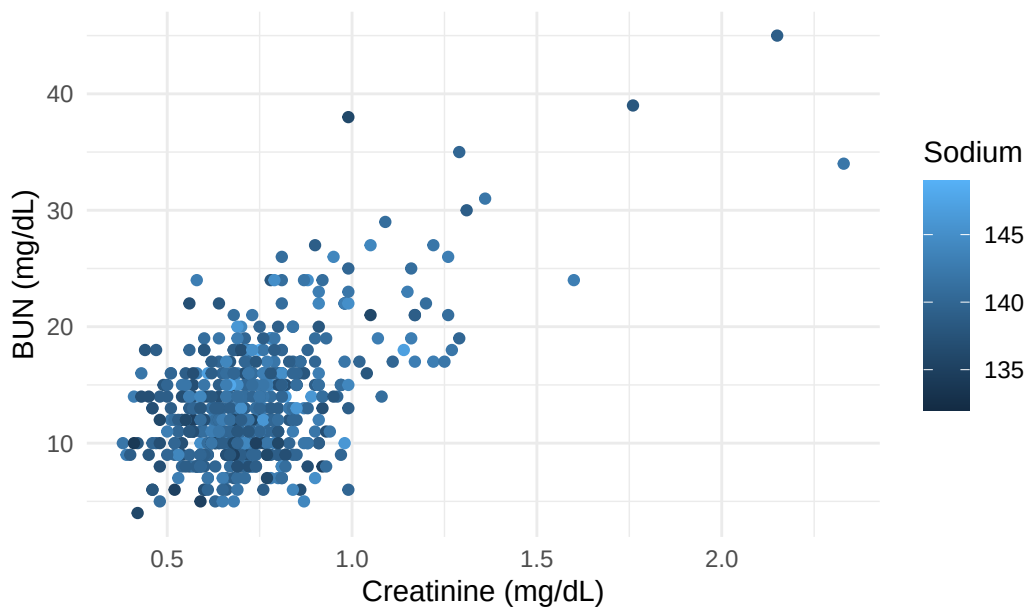
Multiple quantitative values

```
ggplot(chem_wide) +  
  geom_point(mapping = aes(x = creatinine,  
                           y = blood_urea_nitrogen,  
                           color = factor(gender))) +  
  theme_minimal() +  
  labs(x = "Creatinine (mg/dL)", y = "BUN (mg/dL)",  
       title = "Correlation of Creatinine and BUN by Gender",  
       color = "Gender")
```



```
ggplot(filter(chem_wide, pat_type == 'inpatient', gender == "Female",  
              creatinine < 5)) +  
  geom_point(mapping = aes(x = creatinine,  
                           y = blood_urea_nitrogen,  
                           color = sodium)) +  
  theme_minimal() +  
  labs(x = "Creatinine (mg/dL)", y = "BUN (mg/dL)",  
        title = "Correlation for Inpatient Females by Sodium",  
        color = "Sodium")
```


Correlation for Inpatient Females by Sodium



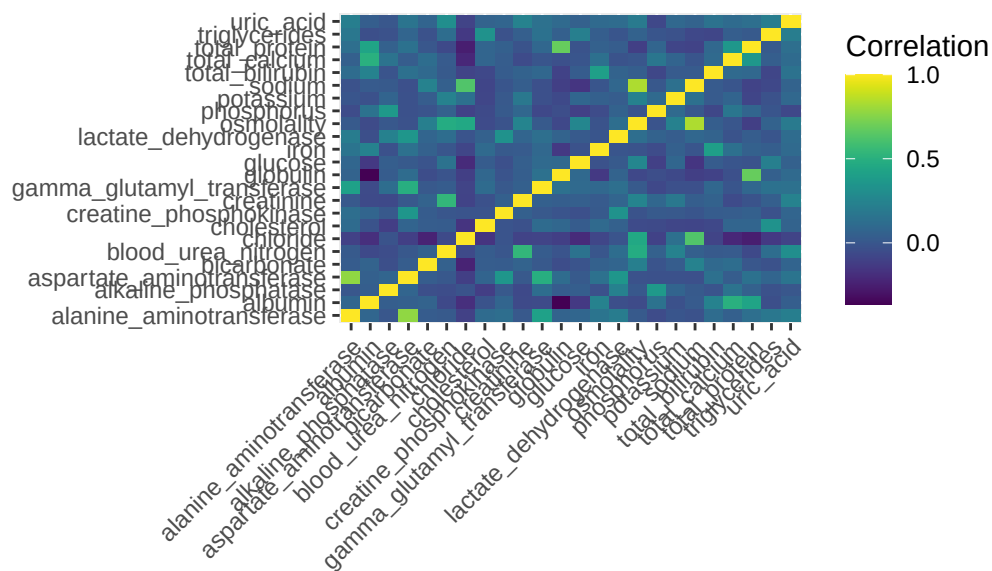
```
df <- select(chem_wide, alanine_aminotransferase:uric_acid)

cor_df <- cor(df) |>
  as_tibble(rownames = "var1") |>
  pivot_longer(
    cols = -var1, # Pivot all columns except 'Var1'
    names_to = "var2", # New column for original column names
    values_to = "correlation") # New column for correlation values

library(viridis)
```

Loading required package: viridisLite

```
ggplot(data = cor_df, mapping = aes(x = var1, y = var2,
                                     fill = correlation)) +
  geom_tile() +
  scale_fill_viridis(discrete=FALSE) +
  labs(x = "", y = "",
       title = "",
       fill = "Correlation") +
  theme(axis.text.x = element_text(angle = 45, vjust = 1, hjust = 1))
```

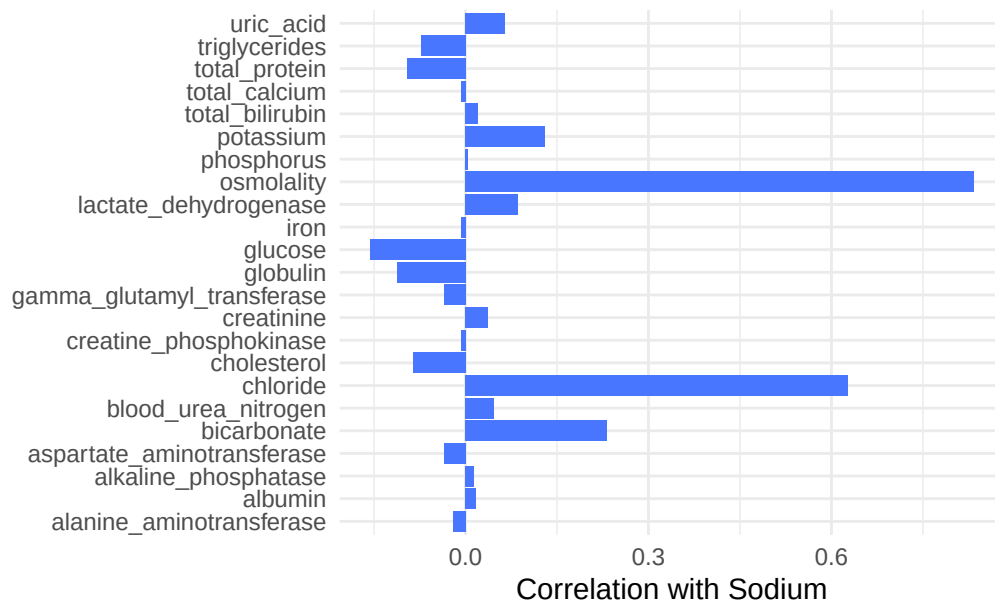


```
# flip x axis labels!!
```

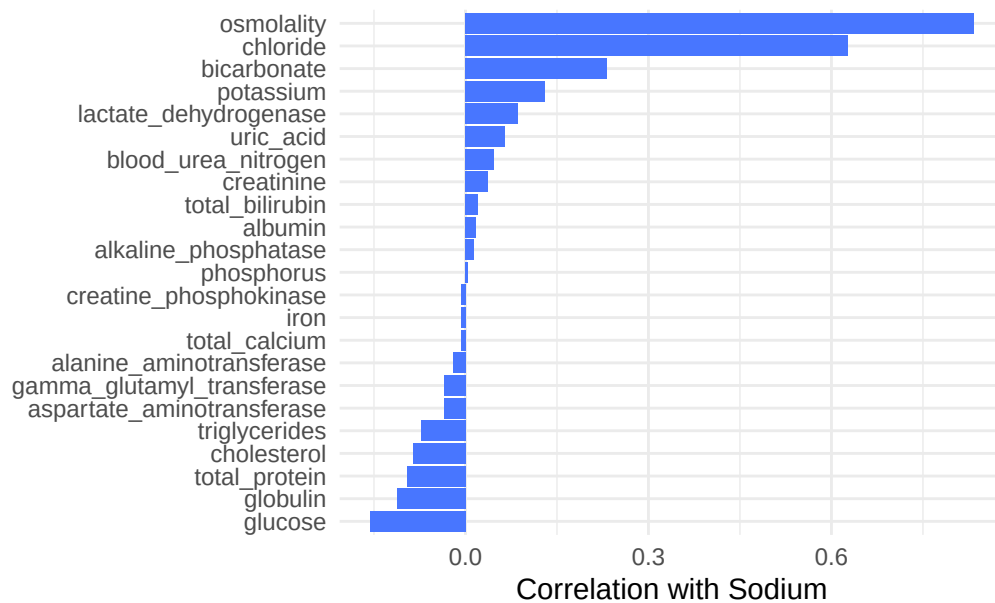
Make comparisons easier

```
sodium_cor <- filter(cor_df, var1 == "sodium", correlation != 1.000)
```

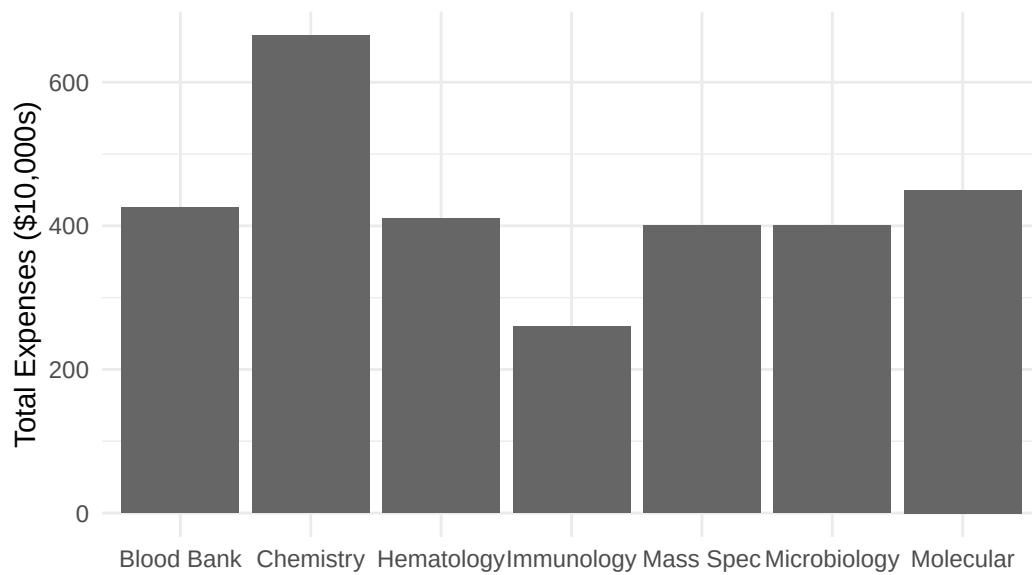
```
ggplot(sodium_cor) +
  geom_bar(mapping = aes(x = correlation, y = var2),
    stat = "identity",
    fill = "royalblue1") +
  theme_minimal() +
  labs(x = "Correlation with Sodium", y = " ", title = "")
```



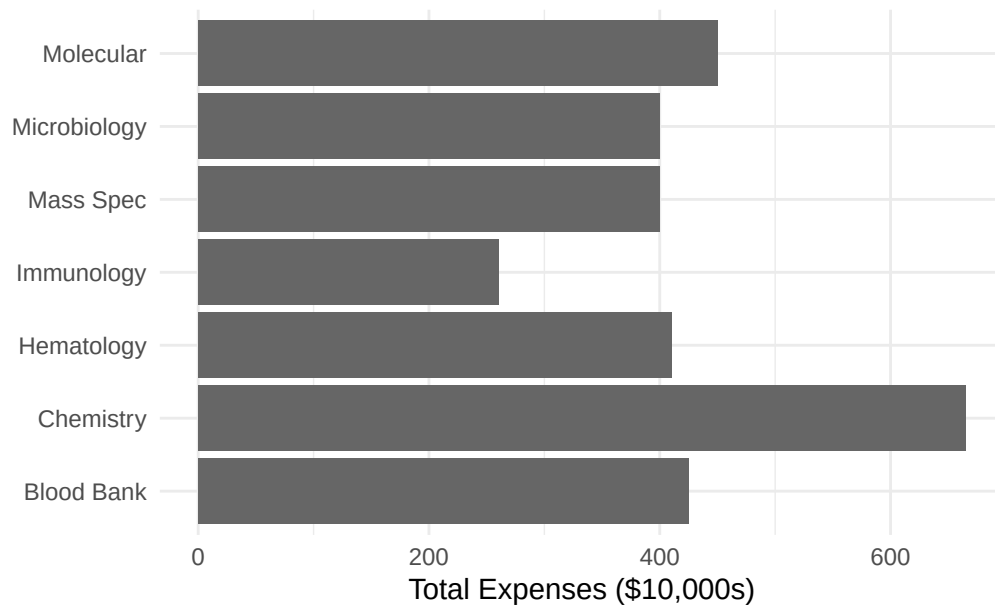
```
ggplot(sodium_cor) +
  geom_bar(mapping = aes(x = correlation,
                        y = fct_reorder(var2, correlation)),
           stat = "identity",
           fill = "royalblue1") +
  theme_minimal() +
  labs(x = "Correlation with Sodium", y = " ",
       title = "")
```



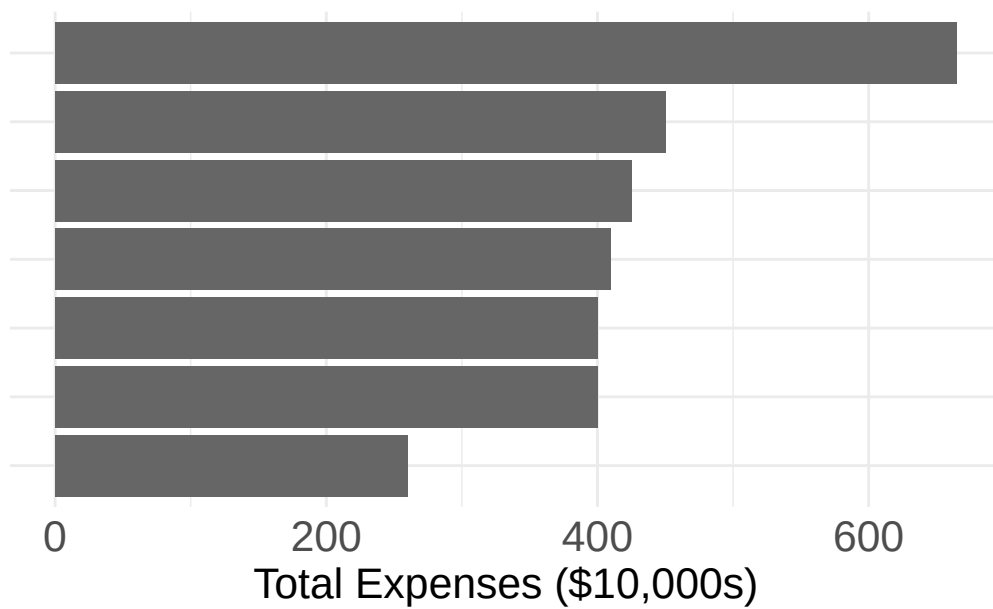
```
ggplot(exp) +
  geom_bar(mapping = aes(x = department, y = total),
           stat = "identity",
           fill = "grey40") +
  theme_minimal() +
  labs(x = "", y = "Total Expenses ($10,000s)",
       title = "")
```



```
ggplot(exp) +  
  geom_bar(mapping = aes(x = department, y = total),  
            stat = "identity",  
            fill = "grey40") +  
  coord_flip() +  
  theme_minimal() +  
  labs(x = "", y = "Total Expenses ($10,000s)",  
        title = "")
```

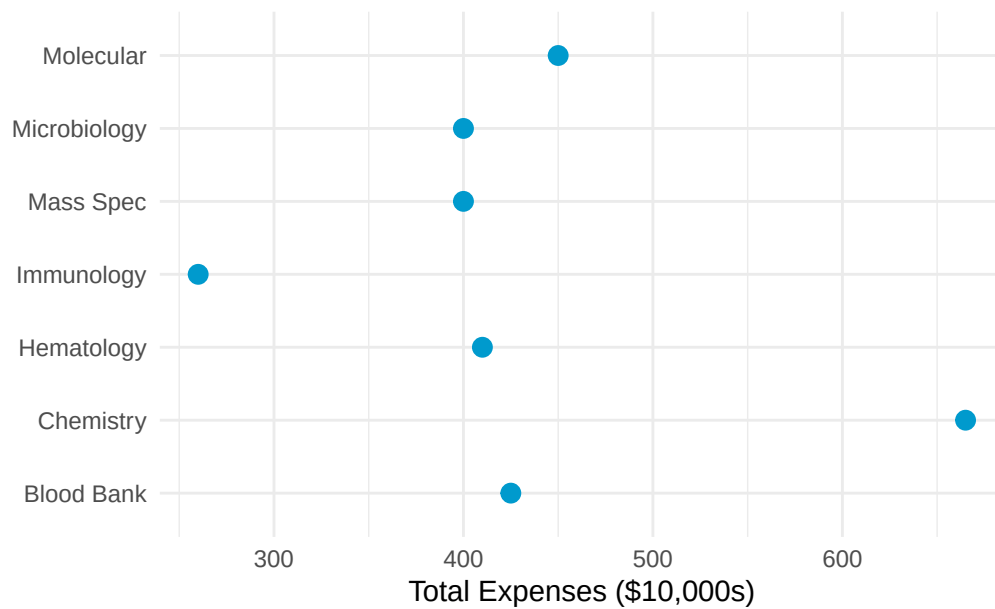


```
ggplot(exp) +  
  geom_bar(mapping = aes(x = fct_reorder(department, total,  
                                         .desc = FALSE),  
                        y = total),  
           stat = "identity",  
           fill = "grey40") +  
  coord_flip() +  
  theme_minimal() +  
  theme(axis.text.x = element_text(size = 16), axis.text.y = element_blank(),  
        axis.title.x = element_text(size = 16)) +  
  labs(x = "", y = "Total Expenses ($10,000s)",  
       title = "")
```



Effective visual encodings

```
ggplot(exp) +  
  geom_point(mapping = aes(x = total, y = department),  
             color = "deepskyblue3",  
             size = 3) +  
  theme_minimal() +  
  labs(x = "Total Expenses ($10,000s)", y = "",  
       title = "")
```



```
ggplot(exp) +  
  geom_point(mapping = aes(x = total,  
                           y = fct_reorder(department, total)),  
             color = "deepskyblue3",  
             size = 3) +  
  #xlim(0, 700) +  
  theme_minimal() +  
  theme(axis.text.x = element_text(size = 16),  
        axis.text.y = element_text(size = 16),  
        axis.title.x = element_text(size = 16)) +  
  labs(x = "Total Expenses ($10,000s)", y = "",  
       title = "")
```